



SRI VASAVI ENGINEERING COLLEGE

(AUTONOMOUS)

(Sponsored by Sri Vasavi Educational Society)

(Approved by AICTE, New Delhi & Recognized by UGC under section 2(f) & 12(B))

(Permanently affiliated to JNTUK, Kakinada, Accredited by NBA and NAAC with 'A' Grade)

Pedatadepalli, **TADEPALLIGUDEM-534 101**, W.G. Dist. (A.P)

Department of Electrical & Electronics Engineering (NBA Accredited)

Format for Abstract

Name of the Course: - Mini Project

Course Code: - V20SIMP01

Name of the Project: - TRANSMISSION LINE FAULT DETECTION SYSTEM.

Name of the Supervisor: - Dr. CHAPPA ANIL KUMAR Ph.D

Name & Roll No.'s of Students: -

21A81A0276	G. YESWANTH
21A81A0233	P. LAVA KUMAR
21A81A0295	P. GAYATHRI
21A81A0287	M. VINOD KUMAR

Abstract

Now a days transmission line protection is a key problem in power transmission line because 85%-87% of power system disturbances are occurring in transmission lines. Identification of fault source is tedious task; fast fault detection can help to protect the equipment before any significant damage of the equipment. The exact fault location can help service man to remove persistent of the faults and locate the areas where the fault occurs regularly, thus reducing the occurrence of fault in transmission line using an Arduino board and the same is transmitted to control center using Arduino device. In this project, we sense the current by placing the conducting wire inside the loop of the current transformer and is given to the protective circuit to avoid the high current, this is given to the Arduino. The fire sensor (An IR receiver) is given to the Arduino to detect the presence of any fire. In case there is a short circuit, the current in the series resistor modifies according to the resistance that modifies with the distance and the load which is connected through a relay to the Arduino is turned OFF when the resistance is below threshold value. In addition, this thesis can also be empowered by using capacitor in the AC circuit to measure the impedance that could also locate the open circuit cable. An LCD display will indicate the status of the transmission lines and buzzer will sound when fault is detected. The SMS alerts will be going to be sent to the corresponding number through GSM Modem.

Supervisor

Project Co-Ordinator

Head of the Department

Department Vision:

- To evolve as a centre of excellence in Electrical and Electronics Engineering that produces graduates of high quality with ethical values.

Department Mission:

- To impart technical knowledge through learner-centric education supplemented with practical exposure.
- To provide opportunities that promote personality development through co-curricular and extra-curricular activities.
- To inculcate human values & team spirit that enables the Electrical and Electronics Engineers to face the future challenges.